# Incident Triage Agent (Agent 1)

*Deep Technical Specification & Operational Manual*

## Executive Overview & Purpose

The Incident Triage Agent is the primary defensive analysis component responsible for processing security logs, network events, and threat signals. It parses unformatted event streams, extracts indicators of compromise (IOCs) using regular expression engines, enriches them against external threat intelligence APIs, and performs LLM-driven classification.

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## 1. Architectural Overview & Workflow

- Position in LangGraph Engine: Entry node for Pipeline B (Incident Response Pipeline) and Pipeline A_THEN_B.
- Input Ingestion: Accepts raw syslog strings, Windows Event Logs (EVTX JSON), firewall logs, NIDS/NIPS alerts (Suricata/Snort), and cloud audit trails.
- State Propagation: Populates state.extracted_iocs and state.triage_report in the central SentinelState data structure.
- Execution Latency: Sub-3 second execution utilizing gemini-2.0-flash with Grafify token compaction.

## 2. IOC Extraction & Regular Expressions Engine

- IPv4 Regex: Matches public IPv4 addresses while automatically filtering private ranges (10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16, 127.0.0.1/8).
- Domain Regex: Matches top-level domains (.com, .net, .org, .io, .ru, .cn, .xyz, etc.) with RFC-compliant validation.
- Cryptographic Hashes: Extracts MD5 (32-hex), SHA-1 (40-hex), and SHA-256 (64-hex) hashes from log messages and file transfer events.
- Deduplication: Automatically deduplicates duplicate IOCs while preserving first-seen and last-seen timestamp metadata.

## 3. Real-Time External API Integrations

- VirusTotal API v3: Asynchronously queries /api/v3/ip_addresses/{ip}, /domains/{domain}, and /files/{hash}. Extracts last_analysis_stats (malicious, suspicious, harmless votes) and threat categories.
- Shodan REST API: Queries /shodan/host/{ip} to retrieve host OS, ISP attribution, hostnames, and active open ports (e.g. 22/SSH, 80/HTTP, 445/SMB, 3389/RDP).
- Fallback & Resilience: Gracefully degrades to local heuristic scoring if API keys are exhausted or external endpoints experience timeout.

## 4. LLM Triage Analysis & Output Schema

- True/False Positive Classification: Evaluates whether log signals represent an authentic security breach or benign network noise.
- Severity Scoring: Assigns CRITICAL, HIGH, MEDIUM, LOW, or INFORMATIONAL severity based on threat impact and asset exposure.
- Schema Contract (TriageAnalysisSchema): Output strictly validated via Pydantic schema containing classification, confidence, attack_pattern, affected_assets, key_findings, and recommended_immediate_actions.